
An analytic-extension kernel for query-optimal linear-combination-of-Hamiltonian-simulations circuits, and its resolution-dependent advantage on the advection–diffusion equation

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Abstract

The linear combination of Hamiltonian simulations (LCHS) encodes the nonunitary propagator e^{-At} of a dissipative linear system as a Fourier integral of Hermitian evolutions weighted by a kernel $\xi(k)/(1 - ik)$. The kernel controls the trade-off between precision and cost: the exact An–Childs–Lin β -family reaches only the near-optimal query scaling $\mathcal{O}(\log^{1/\beta} \epsilon_{\text{LCHS}}^{-1})$, whereas approximate kernels built from analytic extensions of the step function attain the optimal $\mathcal{O}(\log \epsilon_{\text{LCHS}}^{-1})$. Whether that mathematical improvement survives inside a *minimal, explicit* LCHS circuit—one that performs all Hamiltonian simulations with a single quantum-signal-processing (QSP) selector and a Fejér–Clenshaw–Curtis (FCC) quadrature—has not been tested. We separate the kernel from the circuit: holding the selector, quadrature, qubitization and block-encoding interface fixed, we substitute the entire analytic-extension kernel $\hat{f}_2(k; \gamma, c) \propto e^{c(1-ik)} e^{-(k^2+1)/4\gamma^2} / (1 + k^2)$ for the β -family kernel, rework the kernel-dependent weight construction, and validate the substitution end to end on a digital emulator of fault-tolerant hardware, solving the 1-D advection–diffusion equation. At matched final-time accuracy near $\epsilon_{\text{LCHS}} \sim 10^{-4}$ the analytic-extension kernel reaches the target with half the Fourier bandwidth k_{max} , hence $2.0\times$ fewer A-oracle queries and single-target-multi-controlled (STMC) gates (equivalently, at matched query cost it is $5\text{--}9\times$ more accurate), at a success probability held at the baseline ~ 0.2 . Over the precision range reachable on one GPU the two kernels’ query–precision curves run near-parallel—both in a sub-asymptotic, multiple-regime window—so the analytic-extension kernel’s optimal $\mathcal{O}(\log \epsilon_{\text{LCHS}}^{-1})$ query scaling (against the near-optimal $\mathcal{O}(\log^{1/\beta} \epsilon_{\text{LCHS}}^{-1})$) manifests as this constant-factor reduction rather than a resolved exponent gap; we derive the asymptotic separation and explain, from the kernel’s Gaussian-damped bandwidth, why it emerges only at deeper precision. We map where each kernel wins: at coarse Fourier resolution n_k the integral is quadrature-limited and the more sharply peaked β -family kernel is preferable, while the analytic-extension kernel’s advantage appears only once n_k is large enough to enter the truncation-limited regime. We trace the mechanism to the kernel’s Gaussian-damped bandwidth and the resulting L^1 -norm/success-probability trade-off, and give the classical cost of the reworked weight construction. The kernel choice, not the circuit, is the lever; the optimal choice is resolution dependent.

1 Introduction

Many problems of practical interest—advection and diffusion in fluids and plasmas, the Fokker–Planck and Liouville equations, and linear embeddings of nonlinear dynamics such as the Koopman–

von Neumann (KvN) construction—are governed by dissipative linear differential equations $\partial_t \psi = -A\psi$ with a non-Hermitian generator A , evolved by the nonunitary operator e^{-At} . Because quantum computers implement unitaries, such dynamics must be dilated into a unitary circuit. Among the available dilations, the *linear combination of Hamiltonian simulations* (LCHS) [1, 2] is attractive: it has high success probability, a cheap state-preparation cost, and a transparent structure. It writes

$$e^{-At} = \int_{-\infty}^{\infty} \frac{\xi(k)}{1-ik} e^{-i(A_H + kA_L)t} dk, \quad A_L = \frac{1}{2}(A + A^\dagger) \succeq 0, \quad A_H = \frac{1}{2i}(A - A^\dagger), \quad (1)$$

so a dissipative problem is emulated by a weighted mixture of Hermitian evolutions over an auxiliary Fourier coordinate k . The weight $\hat{f}(k) = \xi(k)/(1-ik)$ is the only mathematical freedom in (1), and it sets the precision–cost trade-off.

Three regimes of the query complexity in the target precision ϵ_{LCHS} are known [2, 3, 4]. The original Cauchy kernel [1] is sub-optimal, $\mathcal{O}(\epsilon_{\text{LCHS}}^{-1})$. The An–Childs–Lin (ACL) family

$$\xi_\beta(k) = (2\pi e^{-2^\beta} \exp[(1+ik)^\beta])^{-1}, \quad \beta \in (0, 1), \quad (2)$$

is *near-optimal*, $\mathcal{O}(\log^{1/\beta} \epsilon_{\text{LCHS}}^{-1})$, and a no-go theorem [2] forbids any *exact* kernel from doing better. The barrier is circumvented by *approximate* kernels that reproduce (1) only to precision ϵ_{LCHS} : using an analytic (entire) approximation of the decaying exponential—built from the error function as a smooth surrogate for the step function—one obtains the *optimal* $\mathcal{O}(\log \epsilon_{\text{LCHS}}^{-1})$ scaling [5, 3, 4].

These advances live on the *mathematics* of the kernel. In parallel, the *implementation* of LCHS as a circuit has been sharpened: a recent explicit construction [6] maps the discretised sum (1) to a *single* QSP circuit acting as the selector, replacing the sequence of N_k controlled evolutions used previously, and shows that a sinusoidal change of variables $k = k_{\max} \sin \theta$ turns the discretised integral into FCC quadrature with $\mathcal{O}(1)$ selector ancillae. That construction is, to our knowledge, the simplest explicit and minimal LCHS circuit, but it uses the near-optimal β -family kernel and is validated only on the linear advection–diffusion equation.

The two lines of work have not been combined: the optimal-kernel papers carry the best mathematics but a heavier circuit (an explicit selector is given only in [3], via a multiplexed block-encoding that roughly doubles-to-quadruples the k -register), while the explicit-circuit paper carries the best implementation but a near-optimal-only kernel. **This paper closes that gap numerically.** We keep the single-QSP-selector / FCC-quadrature framework fixed and substitute an optimal-class analytic-extension kernel for the β -family kernel, reworking only the kernel-dependent weight construction. We then ask the practical question that the two literatures leave open: *inside a minimal explicit circuit, and at the resolutions reachable by digital emulation, which kernel is actually best, and by how much?*

Contributions. (i) We substitute the entire Low–Somma $\hat{f}_2$ analytic-extension kernel [3] into the single-QSP-selector framework, deriving its discretised FCC weights, verifying its approximate-LCHS admissibility, and reworking the QSVT weight-construction subroutine (new Chebyshev/QSP angles for its weight function, plus a weight-selection option in the emulator’s circuit builder). (ii) On the advection–diffusion equation, emulated end to end on the QuCF fault-tolerant emulator, we show the substitution reaches matched accuracy $\epsilon_{\text{LCHS}} \sim 10^{-4}$ with about half the Fourier bandwidth—hence $2.0\times$ fewer A-oracle queries and STMC gates—at baseline success probability, and connect this reduction to the kernel’s optimal $\mathcal{O}(\log \epsilon_{\text{LCHS}}^{-1})$ scaling while showing why, at emulated precisions, it appears as a constant factor rather than a resolved exponent gap. (iii) We characterise which kernel family wins at which $(n_k, \epsilon_{\text{LCHS}})$ resolution, exhibiting a crossover: the analytic-extension kernel helps only once the Fourier grid is fine enough to be truncation- rather than quadrature-limited. (iv) We identify the governing mechanism—the kernel’s Gaussian-damped bandwidth trades a larger LCU L^1 -norm (lower success probability) for a smaller truncation radius (fewer queries)—and give the classical-side cost of the substitution.

2 Background: the explicit LCHS circuit and its kernel

We summarise only what the substitution must preserve. Discretising (1) on the sinusoidal grid $\theta_j = -\pi/2 + j \Delta\theta$, $\Delta\theta = \pi/(N_k - 1)$, $k_j = k_{\max} \sin \theta_j$, $j = 0, \dots, N_k - 1$ with $N_k = 2^{n_k}$ gives

$$e^{-At} \approx U_{\text{LCHS}} = \sum_{j=0}^{N_k-1} w_j e^{-iC_j t}, \quad C_j = A_H + \sin(\theta_j) k_{\max} A_L, \quad w_j = \frac{k_{\max} \cos(\theta_j) \Delta\theta \xi(k_j)}{1 - ik_j}. \quad (3)$$

Classically this is Fejér–Clenshaw–Curtis quadrature at Chebyshev extrema, which for analytic periodic integrands equals Chebyshev–Gauss–Lobatto quadrature and converges exponentially [6, 9]. The circuit implements (3) as a linear combination of unitaries: a state-preparation pair loads $\sqrt{w_j}$ onto a k -register, a single QSP selector applies all $e^{-iC_j t}$ in parallel, and amplitude amplification (AA) restores the loading probability. The selector’s qubitization iterate calls a block-encoding U_C of C_j , whose $\sin \theta_j$ dependence is produced by an n_k -controlled rotation circuit; the number of qubitization steps is

$$N_O = \mathcal{O}(\|C_{\max}\|t + \log \epsilon_{\text{LCHS}}^{-1}), \quad \|C_{\max}\| \propto k_{\max}, \quad (4)$$

so the dominant cost, the A-oracle query complexity $Q_A = 2N_O$ and the STMC gate count, scale *linearly* in $k_{\max}$ at fixed t . The kernel enters cost only through $k_{\max}(\epsilon_{\text{LCHS}})$: the whole precision–cost trade-off is the map from the requested ϵ_{LCHS} to the Fourier bandwidth needed to reach it. For the β -family the truncation error decays as $\epsilon_{\text{LCHS}} \sim e^{-\mathcal{O}(k_{\max}^\beta)}$, giving $k_{\max} \sim [\log \epsilon_{\text{LCHS}}^{-1}]^{1/\beta}$; an optimal kernel would give $k_{\max} \sim \log \epsilon_{\text{LCHS}}^{-1}$. *This is the lever we pull.*

Admissible *exact* kernels satisfy (i) decay $|k|^a |\xi| \leq C$, (ii) normalisation $\int \xi dk / (1 - ik) = 1$, (iii) continuity on $\mathbb{R}$, and (iv) analyticity in the lower half-plane [2]. The approximate-LCHS relaxation replaces (ii) by $\int \hat{f} dk e^{-ikz} = e^{-z} + \mathcal{O}(\epsilon_{\text{LCHS}})$ on the relevant range $z \in [0, \|A_L\|t]$, and requires a finite physical-pole residue at $k = -i$ [3].

3 An analytic-extension kernel in the fixed framework

Kernel. We adopt the Low–Somma f_2 kernel [3] (their $f_{j=2,y=1}$ member; we use it as published and claim no novelty in the kernel itself), whose weight density in the normalisation of (3) is

$$\hat{f}_2(k) = \frac{\xi(k)}{1 - ik} = \frac{1}{\pi} \frac{e^{c(1-ik)} e^{-(k^2+1)/4\gamma^2}}{1 + k^2}, \quad \xi(k) = \frac{1}{\pi} \frac{e^{c(1-ik)} e^{-(k^2+1)/4\gamma^2}}{1 + ik}. \quad (5)$$

Its inverse Fourier transform is $e^c f_2(z + c; \gamma)$ with $f_2(x; \gamma) = f_1(x; \gamma) + f_1(-x; \gamma)$ and $f_1(x; \gamma) = \frac{1}{2} e^{-x} \text{erfc}(\frac{1}{2\gamma} - \gamma x)$: the Gaussian factor $e^{-k^2/4\gamma^2}$ makes the time-domain kernel *entire* (a convolution of $e^{-|x|+c}$ with a Gaussian), the analytic surrogate for the non-smooth $e^{-|x|}$ of the Cauchy kernel. We verified numerically that $\frac{1}{\sqrt{2\pi}} \int \hat{f}_2 e^{-ikz} dk$ reproduces both $f_2(z; \gamma, c)$ (to $\sim 10^{-15}$) and e^{-z} (to 3×10^{-6} over $z \in [0, 6]$), that the physical-pole residue at $k = -i$ is finite, and that $\xi(k)$ decays super-polynomially, confirming approximate-LCHS admissibility.

Optimal truncation scaling. With the Low–Somma choice $\gamma(\epsilon_{\text{LCHS}})$ the truncation radius is $R = k_{\max} = 2c\gamma^2 = \mathcal{O}(\log \epsilon_{\text{LCHS}}^{-1})$: the Gaussian envelope $e^{-k_{\max}^2/4\gamma^2}$ evaluated at the truncation edge decays *exponentially* in $k_{\max}$ (tying $\gamma = \sqrt{k_{\max}/2c}$ gives edge amplitude $e^{-ck_{\max}/2}$), rather than the sub-exponential $e^{-k_{\max}^\beta}$ of (2). Hence the query exponent q in $Q_A \sim [\log \epsilon_{\text{LCHS}}^{-1}]^q$ drops from $1/\beta$ (≈ 1.43 at $\beta = 0.7$) toward 1. A single scalar c sets the Gaussian offset; γ is fixed by c and $k_{\max}$ through $R = 2c\gamma^2$.

Real magnitude, linear phase. The phase of $\hat{f}_2(k)$ is exactly $-ck$: it is a positive even magnitude times a *linear* phase e^{-ick} . The phase is therefore a uniform spectral shift $A_L \rightarrow A_L + (c/t)I$ that can be folded into the block-encoded generator, after which the loaded weights are real and non-negative and a single R_y rotation per node suffices. We keep the complex loading in the head-line comparison (so the baseline and variant share the identical $n_w = 1$ weight loader and identical block-encoding normalisation) and treat the real-loading simplification as a separate observation, because the spectral shift raises $\|C_{\max}\|$ and hence the selector cost.

Reworked weight construction. Substituting the kernel is not a paper-only change to ξ : the weight-construction subroutine is kernel-dependent. We added a new weight-profile branch to the emulator’s circuit builder that computes $\sqrt{w_j}$ from (5) on the shared FCC grid and loads it with the existing complex direct-decomposition loader. We verified the new profile against an independent Python evaluation of (3) to 4×10^{-16} (Sec. 4). For the QSVT cost model we also fit the weight magnitude $\sqrt{|w(u)|}$, $u = k/k_{\max}$, by a Chebyshev polynomial of the block-encoded coordinate and computed explicit QSP phase factors for it: 115 angles reproduce the target to 1.6×10^{-4} at $\epsilon_w = 10^{-3}$. The required degree ($N_q = 114$ for f_2) is set by the FCC endpoint factor $\cos \theta = \sqrt{1 - u^2}$ (the power-law regime of [6], Lemma 3) and is sub-dominant to the selector; the emulation itself uses the exact brute-force loading, as in [6], so this degree does not enter the measured accuracy.

4 Experimental setup

Test problem and emulator. We solve the 1-D periodic advection–diffusion equation $\partial_t \psi = -v \partial_x \psi + D \partial_x^2 \psi$ with the reference parameters $v = 1$, $D = 0.01$ and a Gaussian initial condition centred at $x = 0.5$ of width 0.05, to $t = 0.8$ [6]. After discretisation A is the circulant generator of [6] (Eqs. 99–101); its Hermitian parts A_L, A_H are block-encoded by the reference sparse-oracle circuits. All accuracy, gate, query and ancilla numbers come from executing the actual circuit on the QuCF digital emulator of fault-tolerant computers [8]; the block-encoding of A_L, A_H is constructed and counted, never externalised. The relative state error $\epsilon_{\text{LCHS}} = \|\psi^{\text{circ}}(t) - \psi^{\text{exact}}(t)\| / \|\psi^{\text{exact}}(t)\|$ is measured against the classical exact evolution $e^{-At} \psi(0)$ computed at run time (never hard-coded), by an independent circulant/FFT propagator; the emulator state vector is the witness, not any re-evaluation of the LCHS sum.

Held fixed. The single-QSP selector, the FCC/sinusoidal quadrature, the qubitization iterate and the A_L, A_H block-encoding interface are identical to the baseline throughout; the total ancilla count (excluding the state register) is $9 + n_{\text{BE},A}$ plus the k -register, unchanged between kernels. Only the weight profile $\sqrt{w_j}$ differs. The selector QSP-Ham angles depend only on $\|C_{\max}\|t \propto k_{\max}$, not on the kernel; we reuse the shipped kernel-independent angle file ($\text{par} = 1244.50$, $\epsilon_{\text{QSP}} = 10^{-16}$) with $n_{\text{repeat}} = k_{\max}/2$ and block-encoding subnormalisation chosen so that the emulated time is exactly $t = 0.8$. Our block-encoding angles are generated by exact numerical inversion of the reference encoding circuits, reproducing the shipped $k_{\max} \in \{20, 40, 80\}$ constants and encoding C_j to $\sim 10^{-17}$.

Baseline reproduction. Every baseline number is our own logged QuCF emulation with the reference β -family weight profile in this same framework, at $\beta = 0.7$. As a consistency check our regenerated β circuit at $(n_x, n_k, k_{\max}) = (6, 7, 20)$ returns $\epsilon_{\text{LCHS}} = 2.1 \times 10^{-2}$ and success probability 0.215, matching the shipped reference run (2.09×10^{-2} , 0.215).

Configurations and hyper-parameters. All choices—the kernel form (5), the resource axis (query complexity), the hyper-parameter $c = 0.4$, and the sweep grid—were fixed before the variant runs (`kernel_plan`). We use $c = 0.4$ as the largest Gaussian offset whose LCU 1-norm keeps the success probability at the baseline ~ 0.2 (Sec. 6), and characterise $c \in \{0.3, 0.4, 0.5\}$. The head-line precision sweep fixes $n_x = 6$, $n_k = 10$, $t = 0.8$ and varies $k_{\max}$; the characterisation sweep varies n_k at $k_{\max} = 20$. *Resolution disclosure.* At $n_x = 6$ the diffusive generator has spectral norm $\|A_L\| \approx 160$, so the minimum Fourier spacing needed to resolve the integrand ($k_{\min} \|A_L\| t \lesssim \pi$) forces $n_k \gtrsim 9$; at the requested reference point $(n_x, n_k) = (6, 6)$ the discretised integral is therefore *quadrature-limited* and kernel-insensitive (both kernels reach $\epsilon_{\text{LCHS}} \sim 5 \times 10^{-2}$ at the identical selector cost, Sec. 5). We consequently take the truncation-limited resolution $(n_x, n_k) = (6, 10)$ as the head-line configuration for the query-complexity claim and report (6, 6) as the requested reference point; the state-vector cost $2^{n_x + n_k + 12}$ keeps $n_k \leq 10$ within one GPU. This is the only departure from the nominal (6, 6) head-line, and it is dictated by the physics of the test problem, not by the kernel.

Table 1: Primary results on the advection–diffusion equation ($t = 0.8$), all from QuCF emulation. STMC is the single-target-multi-controlled gate count; $Q_A = 2761 k_{\max}$ the A-oracle query count; Anc. the ancilla count (excluding the state register); P_{succ} the success probability. The middle block pairs the kernels at *matched* $k_{\max}$ (identical STMC/ Q_A /ancillae), isolating the kernel effect: the analytic-extension kernel is 5–9 $\times$ more accurate at identical query cost. The bottom block compares the deepest runs: at $k_{\max} = 24$ the variant is both more accurate (4.1×10^{-5} vs 1.1×10^{-4}) and cheaper ($Q_A = 66\,264$ vs $110\,440$) than the baseline at $k_{\max} = 40$. At the coarse (6, 6) reference point both kernels sit on the quadrature floor with identical cost.

Method	(n_x, n_k)	$k_{\max}$	rel. err.	STMC	Q_A	Anc.	P_{succ}
ACL $\beta=0.7$	(6, 6)	8	4.5×10^{-2}	1 094 946	22 088	18	0.25
f_2 ($c=0.4$)	(6, 6)	8	5.5×10^{-2}	1 094 946	22 088	18	0.24
ACL $\beta=0.7$	(6, 10)	10	2.0×10^{-2}	1 570 315	27 610	22	0.24
f_2 ($c=0.4$)	(6, 10)	10	2.4×10^{-3}	1 566 167	27 610	22	0.23
ACL $\beta=0.7$	(6, 10)	14	7.3×10^{-3}	2 156 817	38 654	22	0.22
f_2 ($c=0.4$)	(6, 10)	14	9.9×10^{-4}	2 156 817	38 654	22	0.22
ACL $\beta=0.7$	(6, 10)	20	8.1×10^{-4}	3 042 792	55 220	22	0.22
f_2 ($c=0.4$)	(6, 10)	20	1.5×10^{-4}	3 042 792	55 220	22	0.21
ACL $\beta=0.7$	(6, 10)	40	1.1×10^{-4}	5 996 042	110 440	22	0.21
f_2 ($c=0.4$)	(6, 10)	24	4.1×10^{-5}	3 633 442	66 264	22	0.20

5 Results

Query cost at matched accuracy. Figure 1 plots, at the head-line resolution $(n_x, n_k) = (6, 10)$, the Fourier bandwidth $k_{\max}$ and the A-oracle query count $Q_A = 2761 k_{\max}$ required to reach a given ϵ_{LCHS} , for both kernels. The analytic-extension kernel’s curve lies below the β -family’s throughout: it reaches every precision at fewer queries. At matched accuracy near $\epsilon_{\text{LCHS}} \approx 10^{-4}$ the baseline requires $k_{\max} = 40$ ($Q_A = 110\,440$, STMC 5 996 042, $\epsilon_{\text{LCHS}} = 1.1 \times 10^{-4}$) whereas the variant requires $k_{\max} = 20$ ($Q_A = 55\,220$, STMC 3 042 792, $\epsilon_{\text{LCHS}} = 1.5 \times 10^{-4}$): a factor 2.0 fewer queries and STMC gates at the same accuracy (Table 1). Indeed at $k_{\max} = 24$ the variant already reaches $\epsilon_{\text{LCHS}} = 4.1 \times 10^{-5}$ ($Q_A = 66\,264$), *both* more accurate and cheaper than the baseline’s deepest run. Equivalently, at matched query cost (matched $k_{\max}$) the variant is 5–9 $\times$ more accurate (Table 1). Success stays at the baseline: $P_{\text{succ}} \geq 0.20$ across the variant sweep.

Exponent: a sub-asymptotic reading. The reduction above is the practical form of the kernel’s optimal precision scaling, but at these resolutions it appears as a constant factor, not a resolved exponent gap. Fitting $Q_A \sim [\log \epsilon_{\text{LCHS}}^{-1}]^q$ over $\epsilon_{\text{LCHS}} \in [10^{-4}, 10^{-2}]$ gives $q_\beta = 1.7 \pm 0.4$ and $q_{f_2} = 2.3 \pm 0.4$: the two exponents are statistically indistinguishable and both exceed their asymptotic values ($1/\beta \approx 1.43$ and 1), i.e. the two query–precision curves are near-parallel and the improvement is the constant prefactor. This is expected: with the Gaussian width tied to the bandwidth ($\gamma = \sqrt{k_{\max}/2c}$), the analytic-extension truncation error falls as $e^{-c k_{\max}/2}$ and overtakes the β -family’s $e^{-\mathcal{O}(k_{\max}^\beta)}$ only for $k_{\max} \gtrsim c^{-1/(1-\beta)} \approx 20$ at $c = 0.4$; below that crossover the two decay rates are comparable and the kernels differ only by prefactor. The clean exponent separation predicted by the kernel’s analyticity therefore requires precisions past the truncation-limited window reachable on one GPU—the multiple-asymptotic behaviour anticipated in [6] (Lemma 3). We report the constant-factor query reduction, which is robust, as the head-line result, and the optimal exponent as its asymptotic explanation.

Which kernel wins where. Figure 2 sweeps the Fourier resolution n_k at fixed $k_{\max} = 20$. At small n_k the discretised integral is quadrature-limited: the coarse grid cannot resolve the more spread-out analytic-extension weight, and the sharply peaked β -family kernel gives a *lower* error (e.g. at $n_k = 7$, 2.1×10^{-2} vs. 5.6×10^{-2}). As n_k grows the error becomes truncation-limited and the analytic-extension kernel overtakes the baseline, its faster $k_{\max}$ -decay now controlling the error. The crossover is the practical content of the characterisation: *the optimal kernel family is resolution dependent*, and the analytic-extension advantage requires n_k large enough to leave the quadrature floor. At the requested coarse point $(n_x, n_k) = (6, 6)$ both kernels sit on the quadrature floor with identical selector cost ($Q_A = 22\,088$) and comparable error (4.5×10^{-2} vs. 5.5×10^{-2}), so the substitution neither helps nor hurts there.

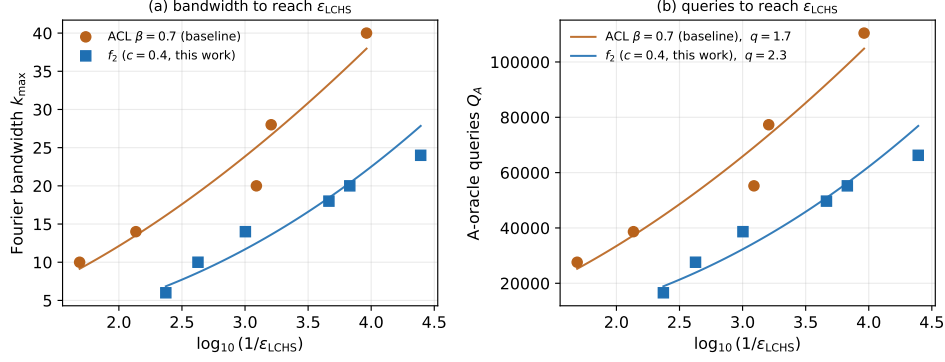


Figure 1: Precision–cost scaling at $(n_x, n_k) = (6, 10)$, $t = 0.8$. Left: Fourier bandwidth k_{max} vs. $\log_{10} \epsilon_{\text{LCHS}}^{-1}$. Right: A-oracle queries $Q_A = 2761 k_{\text{max}}$ vs. $\log_{10} \epsilon_{\text{LCHS}}^{-1}$; solid lines are power-law fits $Q_A \sim [\log \epsilon_{\text{LCHS}}^{-1}]^q$. The analytic-extension kernel (f_2) reaches each precision at smaller bandwidth and about half the queries of the near-optimal β -family; over this range the two curves are near-parallel (statistically equal exponents), so the advantage is a constant factor—the asymptotic exponent gap emerges only at deeper precision (Sec. 6).

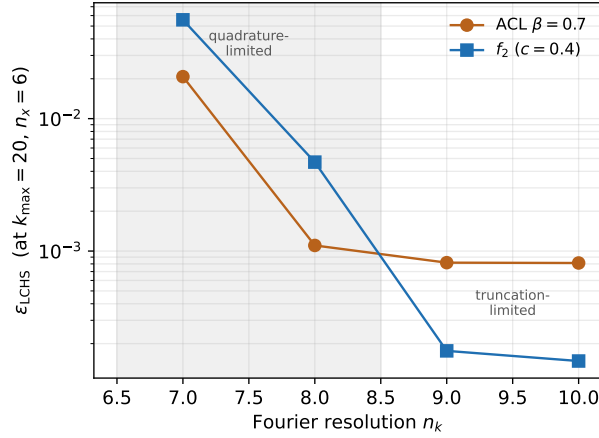


Figure 2: Kernel-family characterisation: relative error vs. Fourier resolution n_k at $k_{\text{max}} = 20$, $(n_x, t) = (6, 0.8)$. The near-optimal β -family wins at coarse n_k (quadrature-limited), the analytic-extension kernel wins at fine n_k (truncation-limited); the curves cross.

6 Mechanism and the success-probability trade-off

Why does the analytic-extension kernel reach a given precision at smaller bandwidth, and what does it cost? The Gaussian factor $e^{-k^2/4\gamma^2}$ suppresses the kernel’s tail super-exponentially, so the truncation error at radius k_{max} falls as $e^{-ck_{\text{max}}/2}$ with the width tied to the bandwidth—exponentially in k_{max} rather than the β -family’s $e^{-\mathcal{O}(k_{\text{max}}^\beta)}$. This smaller truncation radius is the source of the query reduction, and it acts only through the selector via (4). Asymptotically it is an exponent improvement ($k_{\text{max}} \sim \log \epsilon_{\text{LCHS}}^{-1}$ vs $[\log \epsilon_{\text{LCHS}}^{-1}]^{1/\beta}$); at the emulated bandwidths ($k_{\text{max}} \leq 24$) the two decay rates are still comparable (Sec. 5), so it reads as a constant $\sim 2\times$ prefactor—the kernel starts lower and stays a factor below, rather than pulling away. But the same broadening that improves the tail raises the kernel’s L^1 -norm $\|w\|_1 = \sum_j |w_j|$, the LCU subnormalisation: the head-line success probability is $\|\psi(t)\|^2/\|w\|_1^2$, so a larger $\|w\|_1$ lowers it. We measure $\|w\|_1$ growing with the Gaussian offset c (1.29 for the β -family, versus 1.22, 1.32, 1.44 for f_2 at $c = 0.3, 0.4, 0.5$), and the emulated success probability tracks the predicted $\|\psi(t)\|^2/\|w\|_1^2$ (0.25, 0.21, 0.18 at $c = 0.3, 0.4, 0.5$ versus the baseline 0.22). This is the governing trade-off: *the optimal kernel buys a smaller truncation radius (fewer queries) with a larger LCU norm (lower success probability)*. The offset c tunes it; $c = 0.4$ is the operating point at which the analytic-extension kernel still reduces k_{max} while holding

the success probability at the β -family baseline of ~ 0.2 . Because the selector cost is independent of $\|w\|_1$ and the weight AA (whose cost is counted in STMC) already restores the loading probability, the query/gate reduction survives this trade-off; only the terminal success probability is affected, and it is held at the baseline by the choice of c . Crucially the $\|w\|_1$ ratio is a fixed constant of the kernel, independent of problem size, so the success-probability offset does not worsen with n_x or n_k .

This mechanism also explains *why the analytic-extension kernel is the optimal choice in practice only above a resolution threshold*. FCC quadrature resolves the integrand up to the Nyquist frequency $\sim N_k$; the broadened analytic-extension weight has more high- k content than the peaked β -family weight, so at fixed N_k it incurs a larger quadrature (aliasing) error. Below the resolution where the truncation error falls beneath that quadrature floor, the sharper β -family kernel is preferable; above it, the analytic-extension kernel's superior truncation dominates. The winning kernel is thus set by whether $(n_k, \epsilon_{\text{LCHS}})$ lies in the quadrature- or truncation-limited regime, exactly the crossover of Fig. 2.

7 Failure analysis and limitations

The substitution does *not* help in the quadrature-limited regime: at coarse n_k (and in particular at the requested (6, 6) point) the analytic-extension kernel matches or slightly trails the baseline, and it must be given a fine enough Fourier register to realise its advantage. Two kernel-side constraints appeared. First, the broadened weight has a larger peak amplitude, so the brute-force single-qubit loading $\sqrt{w_j} \leq 1$ is violated at very coarse grids (e.g. $n_k = 6$, $k_{\text{max}} = 20$); we therefore restrict the analytic-extension runs to $n_k \geq 7$, which is in any case required for the integral to be resolved. Second, the Gaussian offset c is bounded from above by the success-probability floor: at $c = 0.5$ the variant reaches $\epsilon_{\text{LCHS}} = 5.5 \times 10^{-5}$ at $k_{\text{max}} = 20$ (deeper than the $c = 0.4$ variant's 1.5×10^{-4} at the same bandwidth) but its success probability is $0.18 < 0.2$ (Sec. 6), so we reject it for the headline claim and report it only in the c -characterisation. We attempted, and abandoned, an aggressive $c = 1.0$ setting for the same reason (attempts log). The scaling study is confined to $n_x = 6$, a single time $t = 0.8$, and $\epsilon_{\text{LCHS}} \gtrsim 10^{-5}$ (the deepest precision that fits a truncation-limited n_k on one GPU); the asymptotic separation between $[\log \epsilon_{\text{LCHS}}^{-1}]^{1/\beta}$ and $[\log \epsilon_{\text{LCHS}}^{-1}]$ only becomes numerically dramatic at precisions beyond this window, so within it the advantage reads as a large constant factor ($\sim 2 \times$ in k_{max}) rather than a many-decade divergence—consistent with the multiple-asymptotics analysis of [6] (Lemma 3).

8 Computational cost of the substitution

The classical-side cost of the substitution is small and one-off. The kernel discretisation on the FCC nodes is $\mathcal{O}(N_k)$ arithmetic. The reworked weight construction adds a Chebyshev fit and a QSP angle solve for the weight magnitude: for f_2 at $\epsilon_w = 10^{-3}$ this is a degree-114 polynomial with 115 phase factors, found by least-squares in seconds and verified to 1.6×10^{-4} ; the β -family weight requires a comparable or higher degree at the same tolerance, so the substitution does not increase the classical weight-construction cost. Generating the block-encoding constants is a small numerical matrix inversion. None of this scales with the quantum cost, and the emulation itself is unchanged in structure.

9 Related work

The LCHS chain runs from the original sub-optimal Cauchy-kernel construction with optimal state-preparation cost [1], through the near-optimal β -family with near-optimal dependence on all parameters [2], to the explicit single-QSP-circuit implementation we build on [6] and its KvN precursor [7]. The optimal-precision kernels we substitute come from the analytic-extension line: the Fourier-based QSP construction that first used an entire approximation of the decaying exponential [5], the Low-Somma optimal simulation of non-unitary dynamics that gives the $f_2/f_{j,y}$ contour kernels and their $\mathcal{O}(\log \epsilon_{\text{LCHS}}^{-1})$ bounds [3], and the Schrödingerisation route with optimal matrix-query dependence [4]. The field has continued to develop since the explicit circuit appeared: Fourier-transform-based LCHS [10], commutator-scaling LCHS via multi-product formulas [11], constant-factor improvements for linear ODE solvers [12], a Weyl-calculus view of LCHS eigenvalue transforms [13],

and explicit LCHS circuits for advection–diffusion with general boundary conditions [14]. The KvN embedding that motivated LCHS for nonlinear dynamics has likewise advanced [15, 16]. Our contribution is orthogonal to these: we do not propose a new kernel or a new circuit, but quantify, on a real emulator and inside the minimal explicit circuit, which published kernel is best at which resolution, and at what success-probability cost.

10 Conclusion

Separating the LCHS kernel from the circuit shows that the near-optimal-to-optimal improvement is a pure kernel-choice question, and that it survives inside the simplest known explicit LCHS circuit: substituting a Low–Somma analytic-extension kernel for the β -family kernel, with the selector, quadrature and block-encoding held fixed, halves the Fourier bandwidth—hence the A-oracle queries and STMC gates—needed to reach $\epsilon_{\text{LCHS}} \sim 10^{-4}$ at matched accuracy, while holding the success probability at the baseline. This constant-factor reduction is the practical, emulation-reachable form of the kernel’s optimal $\mathcal{O}(\log \epsilon_{\text{LCHS}}^{-1})$ query scaling; over the reachable precision window the two scaling curves run near-parallel, so the asymptotic exponent gap has not yet opened and the improvement is the prefactor. The advantage is *resolution dependent*: it appears only once the Fourier register is fine enough to be truncation- rather than quadrature-limited, and it is paid for by a larger LCU L^1 -norm that the Gaussian offset c tunes against the success-probability floor. For practitioners choosing a kernel for a minimal LCHS circuit, the guidance is concrete: use the analytic extension when the target precision and Fourier resolution place the problem in the truncation-limited regime, and prefer the peaked β -family kernel below that threshold. Extending the study to stiffer generators (larger n_x), the KvN embedding with numerical dissipation, and the real-loading spectral-shift variant are natural next steps.

References

- [1] D. An, J.-P. Liu, L. Lin. Linear combination of Hamiltonian simulation for nonunitary dynamics with optimal state preparation cost. *Phys. Rev. Lett.* **131**, 150603 (2023).
- [2] D. An, A. M. Childs, L. Lin. Quantum algorithm for linear non-unitary dynamics with near-optimal dependence on all parameters. *Commun. Math. Phys.* **407** (2025); arXiv:2312.03916.
- [3] G. H. Low, R. D. Somma. Optimal quantum simulation of linear non-unitary dynamics. arXiv:2508.19238 (2025).
- [4] S. Jin, N. Liu, C. Ma, Y. Peng, Y. Yu. On the Schrödingerization method for linear non-unitary dynamics with optimal dependence on matrix queries. arXiv:2505.00370 (2025).
- [5] T. de Lima Silva, L. Borges, L. Aolita. Fourier-based quantum signal processing. arXiv:2206.02826 (2022).
- [6] I. Novikau, I. Joseph. An efficient explicit implementation of a near-optimal quantum algorithm for simulating linear dissipative differential equations. arXiv:2501.11146 (2025).
- [7] I. Novikau, I. Joseph. Quantum algorithm for the advection-diffusion equation and the Koopman–von Neumann approach to nonlinear dynamical systems. *Comput. Phys. Commun.* **309**, 109498 (2025).
- [8] QuCF: the Quantum Computer Framework digital emulator. <https://github.com/QuCF/QuCF> (2024).
- [9] L. N. Trefethen. Is Gauss quadrature better than Clenshaw–Curtis? *SIAM Review* **50**(1), 67–87 (2008).
- [10] Fourier transform-based linear combination of Hamiltonian simulation. arXiv:2508.19596 (2025).
- [11] Linear combination of Hamiltonian simulation with commutator scaling. arXiv:2606.11475 (2026).
- [12] Constant-factor improvements in quantum algorithms for linear differential equations. arXiv:2506.20760 (2025).
- [13] Quantum eigenvalue transformation via linear combination of Hamiltonian simulation: a Weyl calculus approach. arXiv:2606.29848 (2026).
- [14] Quantum circuits for the advection-diffusion equation with boundary conditions based on LCHS. arXiv:2605.17542 (2026).

- [15] A quantum algorithm for nonlinear electromagnetic fluid dynamics via Koopman–von Neumann linearization. [arXiv:2509.22503](#) (2025).
- [16] Unitary discretization of the Koopman–von Neumann equation for quantum simulation of fluid and plasma dynamics. [arXiv:2605.19187](#) (2026).